



UGANDA INDUSTRIAL RESEARCH INSTITUTE

Namanve Industrial and Business Park, Mukono District, Uganda

DATABASE DESIGN DOCUMENT

*DESIGN AND DEVELOPMENT OF A WEB BASED
INVENTORY MANAGEMENT SYSTEM
FOR UGANDA INDUSTRIAL RESEARCH INSTITUTE (UIRI)
NAKAWA AND NAMANVE CAMPUSES*

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Project Detail	Description
Programme	Bachelor of Science in Computer Science
Institution	Makerere University, Kampala
Host Organisation	Uganda Industrial Research Institute (UIRI)
Department	Information and Communications Technology (ICT)
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Submitted to	ICT Team, UIRI Namanve
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Document Control

This document defines the database design for the proposed UIRI Inventory Management System. It converts the approved proposal and SRS into a relational data model, data dictionary, relationship rules and normalization evidence for implementation in MySQL or MariaDB.

Table 1 Document Control Details

Item	Description
Document Title	Database Design Document for the UIRI Inventory Management System
Version	1.0
Prepared For	ICT Team, Uganda Industrial Research Institute, Namanve
Prepared By	Keith Paul Kato, Komukama Tracy and Atuhiire Deo Kamate
Primary Design Responsibility	Keith Paul Kato
Design Method	Requirements to conceptual design, logical design and physical relational design
Target DBMS	MySQL or MariaDB
Normalization Target	At least Third Normal Form with additional integrity constraints where necessary

Revision History

Table 2 Revision History

Version	Date	Description	Prepared By
1.0	July-August 2026	Initial database design document prepared after SRS approval	Project Team

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Abbreviations and Acronyms

Table 3 Acronyms and Abbreviations

Acronym	Full Form
DBMS	Database Management System
ERD	Entity Relationship Diagram
EERD	Enhanced Entity Relationship Diagram
FK	Foreign Key
ICT	Information and Communications Technology
IMS	Inventory Management System
MMISDC	Machining, Manufacturing and Industrial Skills Development Centre
PK	Primary Key
RBAC	Role Based Access Control
SRS	Software Requirements Specification
UIRI	Uganda Industrial Research Institute
UOM	Unit of Measure

1.0 Introduction

This Database Design Document presents the data model for the web based Inventory Management System proposed for Uganda Industrial Research Institute. It is prepared after the System Proposal and Software Requirements Specification and is intended to guide the implementation of the database layer, ER diagrams, data dictionary, validation rules and later source code development.

The design responds to the inventory tracking needs identified by the UIRI ICT Team project brief, including item registration, stock movement, supplier records, reports, low stock alerts, user authorization and audit logging [1]. The design also supports the institutional need to report on computers, brands, specifications, year of purchase, acquisition batches, faulty assets, operational assets, serial numbers and location records.

The database has been designed as a normalized relational model. It separates users, campuses, departments, locations, categories, suppliers, item catalogue records, asset instances, stock balances, stock transactions, requisitions, maintenance records, audit logs and report logs. This separation reduces duplication and supports accurate reports.

1.1 Purpose of the Document

The purpose of this document is to define the database structure before coding begins. It provides business rules, entity identification, conceptual ERD, logical ERD, enhanced ERD, data dictionary, normalization evidence and implementation constraints.

1.2 Scope of the Database Design

The scope covers the inventory database for Nakawa and Namanve campuses. The initial pilot may be implemented at UIRI Namanve, but the database includes campus and department tables so that Nakawa can be supported without redesigning the database.

2.0 Database Design Objectives

Table 4 Database Design Objectives

Objective ID	Database Design Objective
DO-01	Support accurate registration of inventory items, categories, suppliers and users.
DO-02	Track both consumable stock and durable fixed assets without mixing their data incorrectly.
DO-03	Allow individual assets such as computers, laboratory equipment and machines to be tracked by serial number, asset tag, purchase year, batch, condition, department and location.
DO-04	Maintain stock balances for consumables and other quantity based items by location.
DO-05	Record every stock movement such as receipt, issue, transfer, return, adjustment and disposal.
DO-06	Support supplier transaction history and acquisition batch reporting.
DO-07	Preserve audit evidence of important user actions for accountability.
DO-08	Provide a reliable database foundation for dashboard summaries and management reports.
DO-09	Reduce redundancy and update anomalies through normalization to at least 3NF.
DO-10	Keep the design implementable using MySQL or MariaDB in the internship prototype environment.

3.0 Design Assumptions and Naming Standards

The following assumptions guide the design. Real inventory records may be incomplete during the first data collection exercise, so the system must support cleaned historical entries and dummy data for demonstration where institutional data is sensitive. Every item belongs to an item category. Every stock movement is performed by a valid system user. Every physical location belongs to a campus and may also be linked to a department. Every fixed asset should have either an asset tag, a serial number or both, depending on the nature of the item.

Table 5 Database Naming Standards

Standard Area	Adopted Standard
Table names	Lowercase plural names, for example users, campuses and inventory_items
Primary keys	Single integer surrogate key ending with _id, for example item_id
Foreign keys	Use the referenced table name in singular form where practical, for example department_id
Dates and times	Use DATE where time is not needed and DATETIME where audit time matters
Quantities	Use DECIMAL for stock quantities to allow units such as litres, metres or kilograms
Text fields	Use VARCHAR for short text and TEXT for long descriptions
Status values	Use lookup tables for important statuses, or controlled values where lookup tables are not necessary
Deletion rules	Do not physically delete records that have transactions. Deactivate or archive them instead

4.0 Business Rules

Table 6 Business Rules

Rule ID	Business Rule
BR-01	One campus can have many departments, but each department belongs to one campus.
BR-02	One campus can have many inventory locations. Each location belongs to one campus and may belong to one department.
BR-03	One user belongs to one role. One role may be assigned to many users.
BR-04	One user may be linked to one department for accountability and report filtering.
BR-05	One category can classify many inventory items. Each inventory item belongs to one category.
BR-06	One inventory item has one unit of measure and one item type.
BR-07	A supplier can supply many items and an item can be supplied by many suppliers. This is resolved using the supplier_items table.
BR-08	A fixed asset item can have many asset instances. Each asset instance represents one physical asset.
BR-09	An asset instance may belong to one acquisition batch, but one acquisition batch can create many asset instances.
BR-10	A consumable or quantity based item can have stock balances in several locations.
BR-11	Each stock transaction must have one transaction type, one item, a quantity, a transaction date and the user who recorded it.
BR-12	A stock transaction may reference a supplier for received stock and may reference a specific asset instance for fixed asset movement.
BR-13	Stock issue cannot reduce a stock balance below zero unless the transaction is an approved correction handled by an authorized user.
BR-14	Each asset instance has one current status such as operational, faulty, under maintenance, retired, disposed or lost.
BR-15	One asset instance can have many maintenance records. Each maintenance record belongs to one asset instance.
BR-16	A requisition can contain many requested items through requisition_lines.
BR-17	Audit logs are append only records and must not be edited by ordinary users.
BR-18	Reports are generated from database records and may be logged in report_logs for accountability.

5.0 Entity Identification

Table 7 Entity Identification Summary

Entity	Purpose
Role	Defines the access level of a user, such as Administrator, Store Manager, Staff User, Management Viewer or Auditor.
UserAccount	Stores system users, login identifiers, department links and role assignments.
Campus	Stores UIRI campus records such as Nakawa and Namanve.
Department	Stores departments or sections under a campus.
Location	Stores physical inventory locations such as stores, laboratories, workshops, rooms and offices.
InventoryCategory	Groups items into categories such as ICT Equipment, Laboratory Equipment and Consumables.
UnitOfMeasure	Defines units such as pieces, boxes, litres, kilograms and metres.
ItemType	Classifies whether an item is a consumable, fixed asset or both.
InventoryItem	Stores the catalogue record for each item type.
Supplier	Stores supplier identification and contact information.
SupplierItem	Resolves the many to many relationship between suppliers and inventory items.
AcquisitionBatch	Stores batch, purchase reference and receipt information for assets or stock received together.
AssetStatus	Stores controlled asset condition values.
AssetInstance	Stores individual durable assets with serial numbers, asset tags and current status.
StockBalance	Stores current item quantity by location.
TransactionType	Stores transaction categories such as stock in, stock out, transfer, return, adjustment and disposal.
StockTransaction	Stores historical stock movement records.
Requisition	Stores inventory requests made by staff or departments.
RequisitionLine	Stores each item requested under a requisition.
MaintenanceRecord	Stores fault, repair and maintenance history for assets.
AuditLog	Stores important user actions for accountability.
ReportLog	Stores report generation events where report access needs to be audited.

6.0 Conceptual Database Design

The conceptual model focuses on the main business objects and their relationships without exposing every physical attribute. It helps stakeholders understand the database before the design is converted into tables and foreign keys.

Figure 1: Conceptual ERD for the UURI Inventory Management System

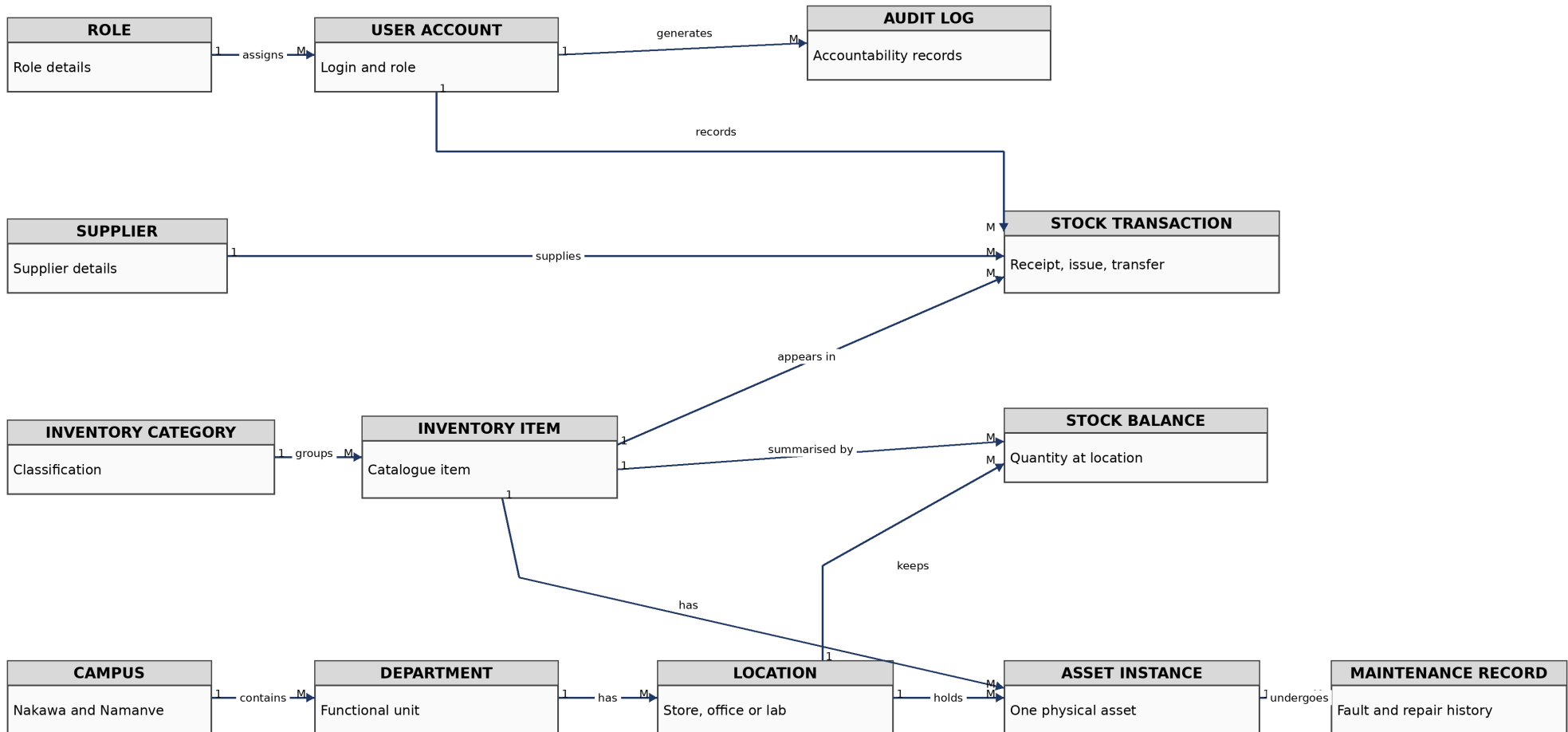


Figure 1 Conceptual ERD for the UURI Inventory Management System

7.0 Logical Database Design

The logical database design converts the conceptual entities into relational tables. The design uses primary keys, foreign keys and relationship constraints to preserve data integrity. For readability, the logical ERD is presented in three Visio-style views: master data, stock movement and asset tracking, and requisitions with audit records.

Figure 2: Logical ERD for Master Data and Inventory Catalogue

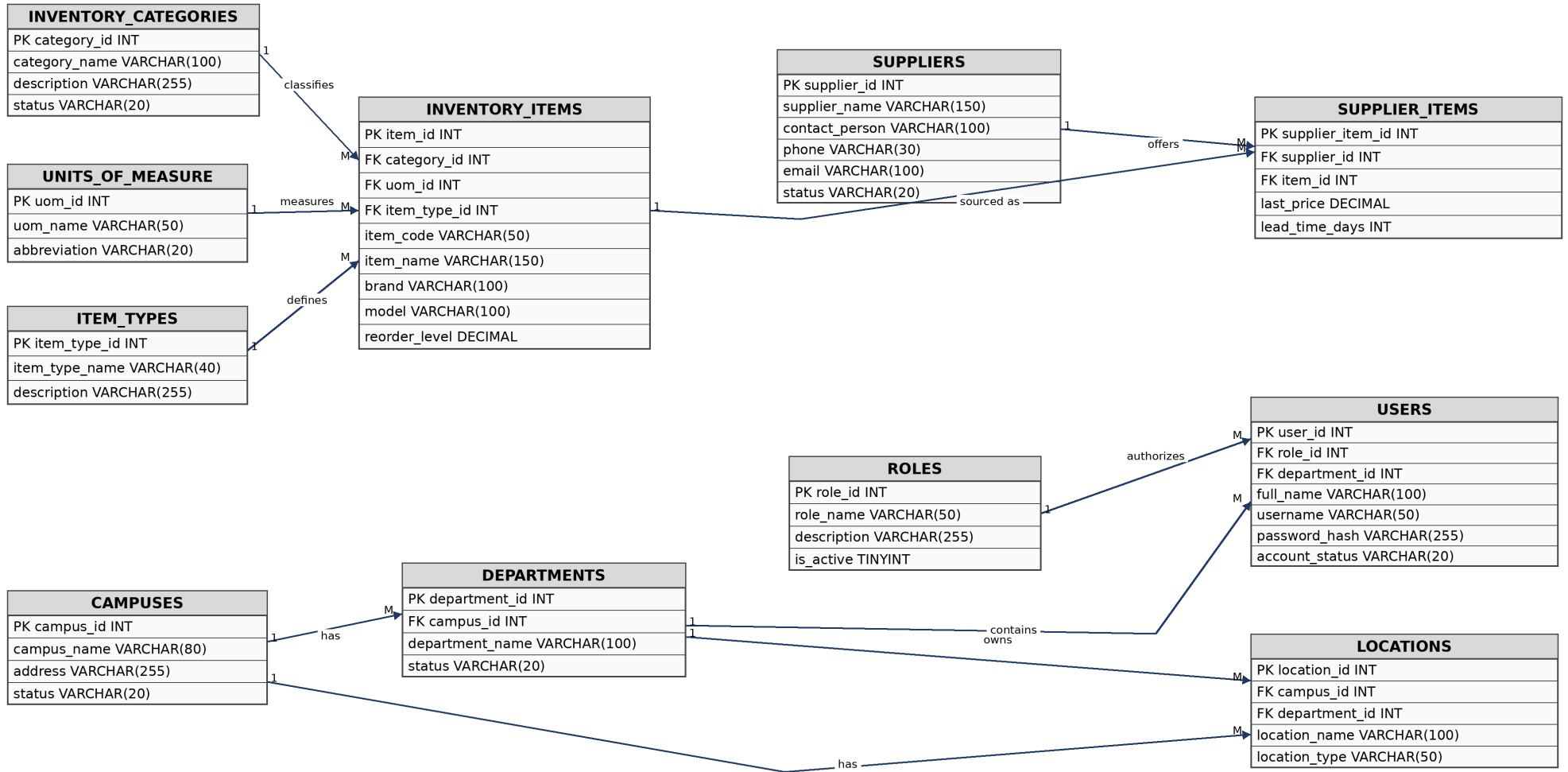


Figure 2 Logical ERD for Master Data and Inventory Catalogue

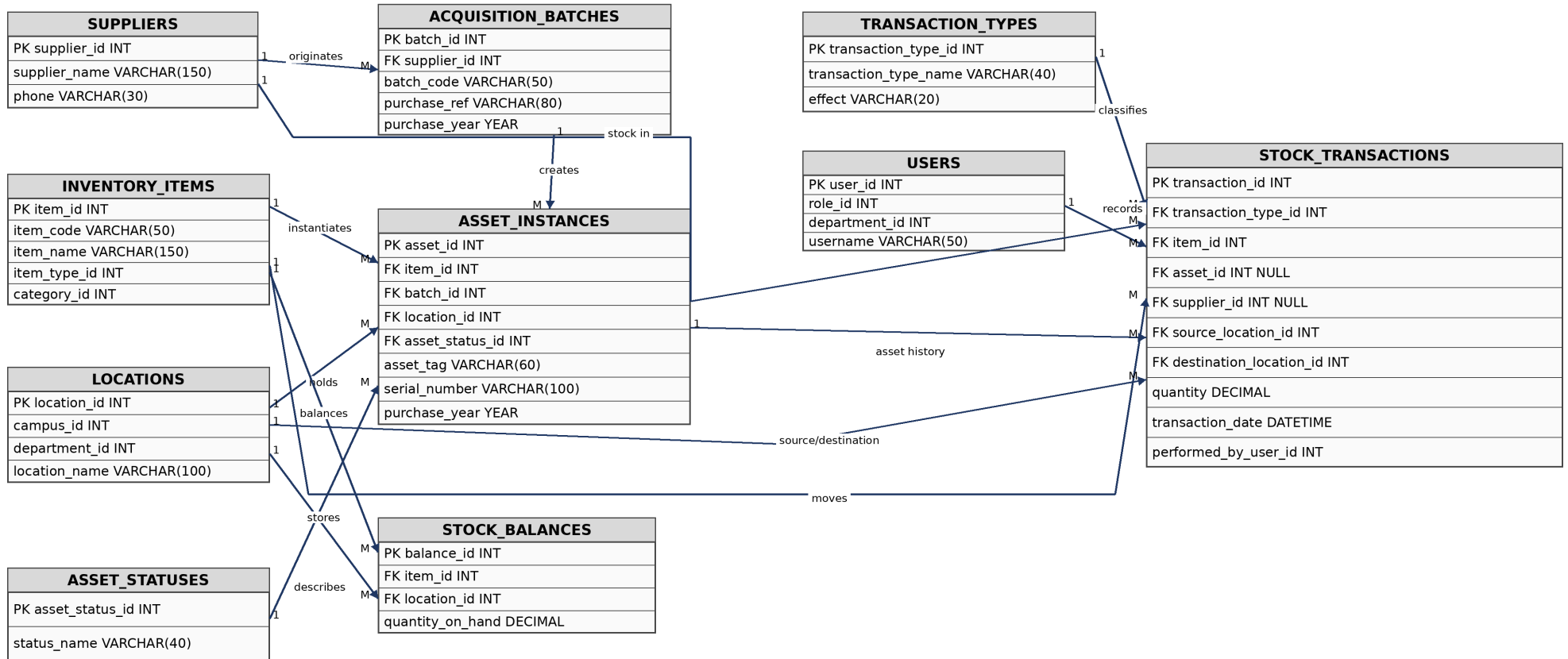
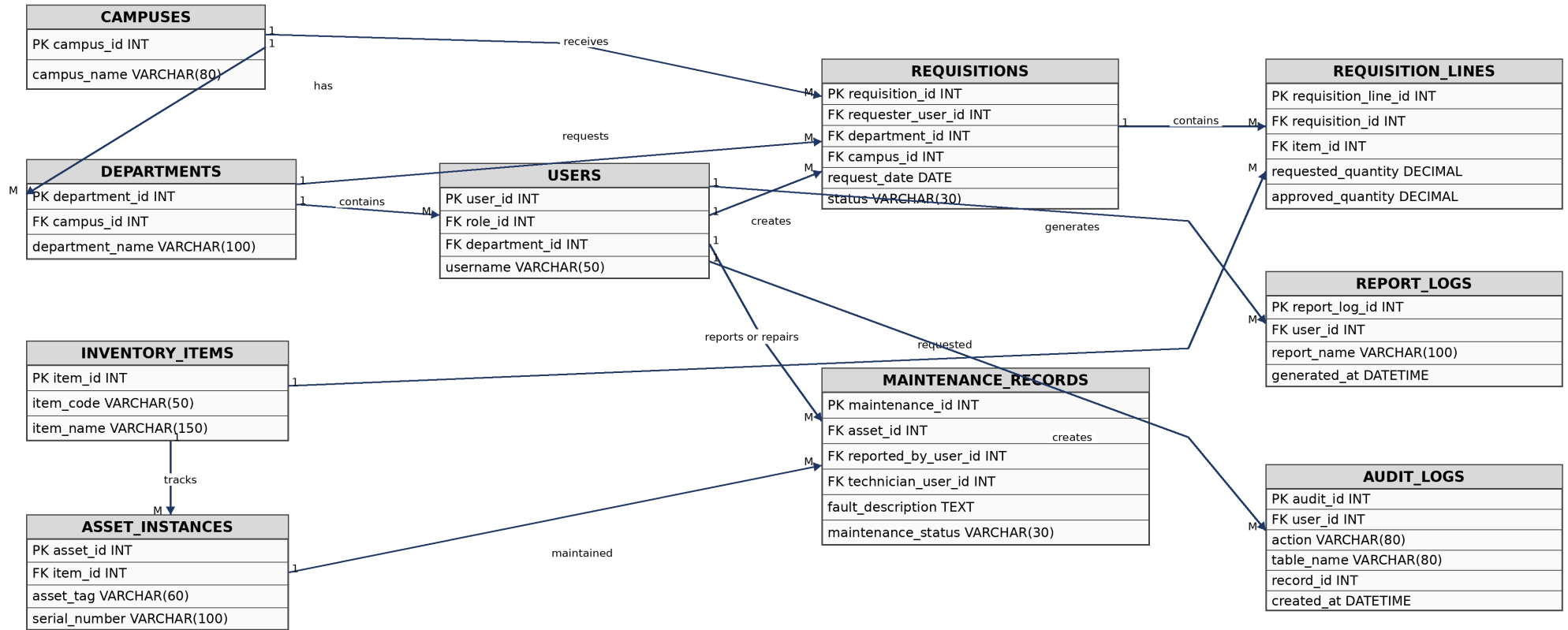
Figure 3: Logical ERD for Stock Movement and Asset Tracking*Figure 3 Logical ERD for Stock Movement and Asset Tracking*

Figure 4: Logical ERD for Requisitions, Maintenance and Audit Records*Figure 4 Logical ERD for Requisitions, Maintenance and Audit Records*

8.0 Enhanced Entity Relationship Design

The enhanced ERD clarifies a critical design decision for UIRI. The system must not treat all items as simple quantities. Consumables are quantity tracked, while fixed assets are individually tracked through asset_instances. This distinction is necessary for reports such as computers in Mechatronics, assets by serial number, faulty equipment and acquisition batches.

Figure 5: Enhanced ERD Showing Item Specialization

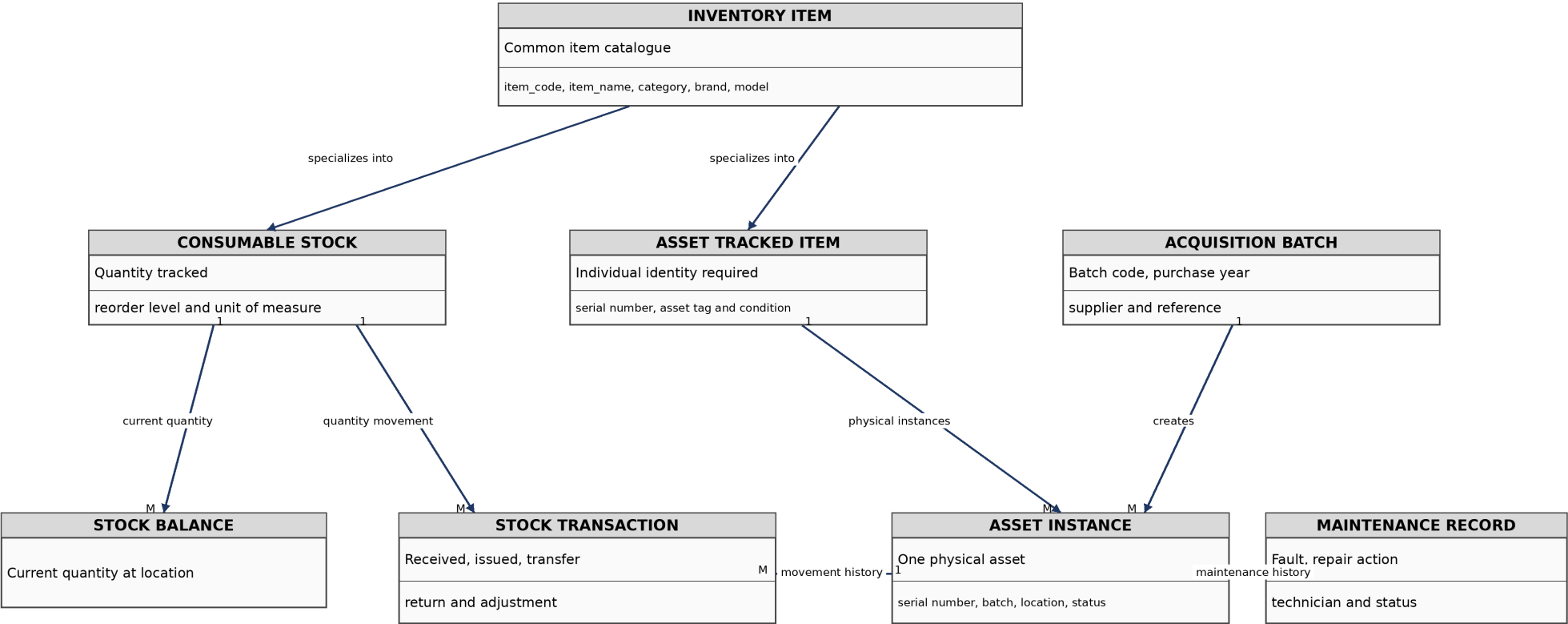


Figure 5 Enhanced ERD Showing Item Specialization

9.0 Data Dictionary

The data dictionary defines each proposed table, field, data type, key constraint and field purpose. Data types are specified for MySQL or MariaDB implementation and may be slightly adjusted during physical implementation where justified by testing.

9.1 roles

Table 8 Data Dictionary for roles

Field	Data Type	Key or Constraint	Description
role_id	INT	PK, AI, NN	Unique role identifier
role_name	VARCHAR(50)	UNIQUE, NN	Role name such as Administrator or Store Manager
description	VARCHAR(255)	NULL	Role description
is_active	TINYINT	NN, default 1	Indicates whether the role is active
created_at	DATETIME	NN	Date and time the role was created

9.2 users

Table 9 Data Dictionary for users

Field	Data Type	Key or Constraint	Description
user_id	INT	PK, AI, NN	Unique user identifier
role_id	INT	FK roles.role_id, NN	Role assigned to the user
department_id	INT	FK departments.department_id, NULL	User department where applicable
full_name	VARCHAR(100)	NN	Full name of the user
username	VARCHAR(50)	UNIQUE, NN	Login username
email	VARCHAR(100)	UNIQUE, NULL	User email address
phone	VARCHAR(30)	NULL	User phone contact
password_hash	VARCHAR(255)	NN	Hashed password value
account_status	VARCHAR(20)	NN	Active, inactive or locked
created_at	DATETIME	NN	Account creation timestamp
updated_at	DATETIME	NULL	Last account update timestamp

9.3 campuses

Table 10 Data Dictionary for campuses

Field	Data Type	Key or Constraint	Description
campus_id	INT	PK, AI, NN	Unique campus identifier
campus_name	VARCHAR(80)	UNIQUE, NN	Campus name such as Namanve or Nakawa
address	VARCHAR(255)	NULL	Campus address
status	VARCHAR(20)	NN	Active or inactive

9.4 departments

Table 11 Data Dictionary for departments

Field	Data Type	Key or Constraint	Description
department_id	INT	PK, AI, NN	Unique department identifier
campus_id	INT	FK campuses.campus_id, NN	Campus to which the department belongs
department_name	VARCHAR(100)	NN	Department or section name
description	VARCHAR(255)	NULL	Department description
status	VARCHAR(20)	NN	Active or inactive

9.5 locations

Table 12 Data Dictionary for locations

Field	Data Type	Key or Constraint	Description
location_id	INT	PK, AI, NN	Unique physical location identifier
campus_id	INT	FK campuses.campus_id, NN	Campus where the location is found
department_id	INT	FK departments.department_id, NULL	Department associated with the location
location_name	VARCHAR(100)	NN	Room, office, store or workshop name
location_type	VARCHAR(50)	NULL	Store, laboratory, workshop, office or room
description	VARCHAR(255)	NULL	Additional location notes
status	VARCHAR(20)	NN	Active or inactive

9.6 inventory_categories

Table 13 Data Dictionary for inventory_categories

Field	Data Type	Key or Constraint	Description
category_id	INT	PK, AI, NN	Unique category identifier
category_name	VARCHAR(100)	UNIQUE, NN	Inventory category name
description	VARCHAR(255)	NULL	Category description
status	VARCHAR(20)	NN	Active or inactive

9.7 units_of_measure

Table 14 Data Dictionary for units_of_measure

Field	Data Type	Key or Constraint	Description
uom_id	INT	PK, AI, NN	Unique unit identifier
uom_name	VARCHAR(50)	UNIQUE, NN	Full unit name
abbreviation	VARCHAR(20)	UNIQUE, NN	Short unit label such as pcs or kg

9.8 item_types

Table 15 Data Dictionary for item_types

Field	Data Type	Key or Constraint	Description
item_type_id	INT	PK, AI, NN	Unique item type identifier
item_type_name	VARCHAR(40)	UNIQUE, NN	Consumable, Fixed Asset or Both
description	VARCHAR(255)	NULL	Explains how items of this type are tracked

9.9 inventory_items

Table 16 Data Dictionary for inventory_items

Field	Data Type	Key or Constraint	Description
item_id	INT	PK, AI, NN	Unique item catalogue identifier
category_id	INT	FK inventory_categories.category_id, NN	Item category
uom_id	INT	FK units_of_measure.uom_id, NN	Unit of measure
item_type_id	INT	FK item_types.item_type_id, NN	Tracking type
item_code	VARCHAR(50)	UNIQUE, NN	Internal item code
item_name	VARCHAR(150)	NN	Item name
brand	VARCHAR(100)	NULL	Brand name where applicable
model	VARCHAR(100)	NULL	Model where applicable
specification	TEXT	NULL	Technical specification
description	TEXT	NULL	General item description
reorder_level	DECIMAL(12,2)	NN, default 0	Minimum stock level
image_path	VARCHAR(255)	NULL	Stored item image path
is_active	TINYINT	NN, default 1	Indicates whether item is active
created_at	DATETIME	NN	Creation timestamp

9.10 suppliers

Table 17 Data Dictionary for suppliers

Field	Data Type	Key or Constraint	Description
supplier_id	INT	PK, AI, NN	Unique supplier identifier
supplier_name	VARCHAR(150)	NN	Supplier business name
contact_person	VARCHAR(100)	NULL	Main contact person
phone	VARCHAR(30)	NULL	Supplier phone contact
email	VARCHAR(100)	NULL	Supplier email
address	VARCHAR(255)	NULL	Supplier address
status	VARCHAR(20)	NN	Active or inactive

9.11 supplier_items

Table 18 Data Dictionary for supplier_items

Field	Data Type	Key or Constraint	Description
supplier_item_id	INT	PK, AI, NN	Unique supplier item identifier
supplier_id	INT	FK suppliers.supplier_id, NN	Supplier
item_id	INT	FK inventory_items.item_id, NN	Item supplied
preferred_status	VARCHAR(20)	NULL	Preferred or alternate supplier indicator
last_price	DECIMAL(12,2)	NULL	Last known price
lead_time_days	INT	NULL	Expected supply lead time
unique_supplier_item	CONSTRAINT	UNIQUE supplier_id, item_id	Prevents duplicate supplier item links

9.12 acquisition_batches

Table 19 Data Dictionary for acquisition_batches

Field	Data Type	Key or Constraint	Description
batch_id	INT	PK, AI, NN	Unique batch identifier
supplier_id	INT	FK suppliers.supplier_id, NULL	Supplier linked to the batch
batch_code	VARCHAR(50)	UNIQUE, NN	Batch or acquisition code
purchase_ref	VARCHAR(80)	NULL	Purchase order or receipt reference
received_date	DATE	NULL	Date the batch was received
purchase_year	YEAR	NULL	Year of purchase
notes	TEXT	NULL	Additional batch notes

9.13 asset_statuses

Table 20 Data Dictionary for asset_statuses

Field	Data Type	Key or Constraint	Description
asset_status_id	INT	PK, AI, NN	Unique status identifier
status_name	VARCHAR(40)	UNIQUE, NN	Operational, faulty, under maintenance, retired, disposed or lost
description	VARCHAR(255)	NULL	Status explanation

9.14 asset_instances

Table 21 Data Dictionary for asset_instances

Field	Data Type	Key or Constraint	Description
asset_id	INT	PK, AI, NN	Unique physical asset identifier
item_id	INT	FK inventory_items.item_id, NN	Catalogue item represented by the asset
batch_id	INT	FK acquisition_batches.batch_id, NULL	Acquisition batch
location_id	INT	FK locations.location_id, NN	Current physical location
department_id	INT	FK departments.department_id, NULL	Current responsible department
custodian_user_id	INT	FK users.user_id, NULL	Current custodian where applicable
asset_tag	VARCHAR(60)	UNIQUE, NN	Internal asset tag
serial_number	VARCHAR(100)	UNIQUE, NULL	Manufacturer serial number
purchase_year	YEAR	NULL	Purchase year
purchase_date	DATE	NULL	Purchase date
acquisition_cost	DECIMAL(12,2)	NULL	Recorded cost if available
asset_status_id	INT	FK asset_statuses.asset_status_id, NN	Current condition or status
warranty_expiry_date	DATE	NULL	Warranty expiry where applicable
remarks	TEXT	NULL	Additional asset notes

9.15 stock_balances

Table 22 Data Dictionary for stock_balances

Field	Data Type	Key or Constraint	Description
balance_id	INT	PK, AI, NN	Unique balance row identifier
item_id	INT	FK inventory_items.item_id, NN	Item whose quantity is held
location_id	INT	FK locations.location_id, NN	Location where quantity is held
quantity_on_hand	DECIMAL(12,2)	NN, default 0	Current available quantity
reorder_level_override	DECIMAL(12,2)	NULL	Location-specific reorder threshold
last_updated_at	DATETIME	NN	Last balance update timestamp
unique_item_location	CONSTRAINT	UNIQUE item_id, location_id	Prevents duplicate balance rows

9.16 transaction_types

Table 23 Data Dictionary for transaction_types

Field	Data Type	Key or Constraint	Description
transaction_type_id	INT	PK, AI, NN	Unique transaction type identifier
transaction_type_name	VARCHAR(40)	UNIQUE, NN	Stock In, Stock Out, Transfer, Return, Adjustment or Disposal
effect	VARCHAR(20)	NN	Increase, decrease, move or no balance
description	VARCHAR(255)	NULL	Explanation of transaction effect

9.17 stock_transactions

Table 24 Data Dictionary for stock_transactions

Field	Data Type	Key or Constraint	Description
transaction_id	INT	PK, AI, NN	Unique stock transaction identifier
transaction_type_id	INT	FK transaction_types.transaction_type_id, NN	Type of stock movement
item_id	INT	FK inventory_items.item_id, NN	Item being moved
asset_id	INT	FK asset_instances.asset_id, NULL	Specific asset moved where applicable
supplier_id	INT	FK suppliers.supplier_id, NULL	Supplier for received stock
source_location_id	INT	FK locations.location_id, NULL	Source location
destination_location_id	INT	FK locations.location_id, NULL	Destination location
quantity	DECIMAL(12,2)	NN	Quantity moved
reference_no	VARCHAR(80)	NULL	Voucher, receipt or request reference
transaction_date	DATETIME	NN	Transaction date and time
performed_by_user_id	INT	FK users.user_id, NN	User who recorded the transaction
approved_by_user_id	INT	FK users.user_id, NULL	User who approved the transaction
remarks	TEXT	NULL	Transaction notes

9.18 requisitions

Table 25 Data Dictionary for requisitions

Field	Data Type	Key or Constraint	Description
requisition_id	INT	PK, AI, NN	Unique requisition identifier
requester_user_id	INT	FK users.user_id, NN	User who made the request
department_id	INT	FK departments.department_id, NN	Requesting department
campus_id	INT	FK campuses.campus_id, NN	Requesting campus
request_date	DATE	NN	Date of request
required_date	DATE	NULL	Date item is needed
status	VARCHAR(30)	NN	Pending, approved, rejected, issued or cancelled
purpose	TEXT	NULL	Reason for the request
approved_by_user_id	INT	FK users.user_id, NULL	Approving user

9.19 requisition_lines

Table 26 Data Dictionary for requisition_lines

Field	Data Type	Key or Constraint	Description
requisition_line_id	INT	PK, AI, NN	Unique requisition line identifier
requisition_id	INT	FK requisitions.requisition_id, NN	Parent requisition
item_id	INT	FK inventory_items.item_id, NN	Requested item
requested_quantity	DECIMAL(12,2)	NN	Quantity requested
approved_quantity	DECIMAL(12,2)	NULL	Quantity approved
issued_quantity	DECIMAL(12,2)	NULL	Quantity issued
line_status	VARCHAR(30)	NN	Pending, approved, issued or rejected
remarks	TEXT	NULL	Line notes

9.20 maintenance_records

Table 27 Data Dictionary for maintenance_records

Field	Data Type	Key or Constraint	Description
maintenance_id	INT	PK, AI, NN	Unique maintenance identifier
asset_id	INT	FK asset_instances.asset_id, NN	Asset maintained
reported by user id	INT	FK users.user_id, NN	User who reported the fault
technician_user_id	INT	FK users.user_id, NULL	Technician or responsible user
reported_date	DATE	NN	Date fault was reported
maintenance_date	DATE	NULL	Date maintenance was performed
fault_description	TEXT	NN	Problem description
action_taken	TEXT	NULL	Repair or maintenance action
cost	DECIMAL(12,2)	NULL	Maintenance cost if recorded
resulting_status_id	INT	FK asset_statuses.asset_status_id, NULL	Status after maintenance
maintenance_status	VARCHAR(30)	NN	Open, in progress, completed or deferred

9.21 audit_logs

Table 28 Data Dictionary for audit_logs

Field	Data Type	Key or Constraint	Description
audit_id	INT	PK, AI, NN	Unique audit log identifier
user_id	INT	FK users.user_id, NULL	User who performed the action
action	VARCHAR(80)	NN	Action performed
table name	VARCHAR(80)	NN	Affected table
record id	INT	NULL	Affected record identifier
old_value	TEXT	NULL	Previous value where practical
new_value	TEXT	NULL	New value where practical
ip_address	VARCHAR(45)	NULL	User IP address
created_at	DATETIME	NN	Audit timestamp

9.22 report_logs

Table 29 Data Dictionary for report_logs

Field	Data Type	Key or Constraint	Description
report_log_id	INT	PK, AI, NN	Unique report log identifier
user_id	INT	FK users.user_id, NN	User who generated the report
report_name	VARCHAR(100)	NN	Report generated
filter_summary	TEXT	NULL	Filters applied
generated_at	DATETIME	NN	Report generation timestamp
output_format	VARCHAR(20)	NULL	HTML, PDF, Excel or print view

10.0 Relationship Summary

Table 30 Relationship Summary

Parent Table	Child Table	Cardinality	Meaning
roles	users	1:M	One role can be assigned to many users.
campuses	departments	1:M	One campus contains many departments.
campuses	locations	1:M	One campus contains many physical locations.
departments	locations	1:M	One department may control many locations.
departments	users	1:M	One department may have many system users.
inventory_categories	inventory_items	1:M	One category groups many items.
units_of_measure	inventory_items	1:M	One unit can measure many items.
item_types	inventory_items	1:M	One item type applies to many items.
suppliers	supplier_items	1:M	One supplier can be linked to many item supply records.
inventory_items	supplier_items	1:M	One item can be supplied by many suppliers.
suppliers	acquisition_batches	1:M	One supplier can supply many acquisition batches.
inventory_items	asset_instances	1:M	One item can have many physical asset instances.
acquisition_batches	asset_instances	1:M	One batch can contain many assets.
asset_statuses	asset_instances	1:M	One status can describe many assets.
locations	asset_instances	1:M	One location can hold many assets.
inventory_items	stock_balances	1:M	One item can have balances in many locations.
locations	stock_balances	1:M	One location can hold balances for many

Parent Table	Child Table	Cardinality	Meaning
			items.
transaction_types	stock_transactions	1:M	One transaction type can classify many stock transactions.
inventory_items	stock_transactions	1:M	One item can appear in many stock transactions.
asset_instances	stock_transactions	1:M optional	One asset can have many movement records where asset level tracking is used.
users	stock_transactions	1:M	One user can record many stock transactions.
requisitions	requisition_lines	1:M	One requisition can contain many item lines.
inventory_items	requisition_lines	1:M	One item can appear on many requisition lines.
asset_instances	maintenance_records	1:M	One asset can have many maintenance records.
users	audit_logs	1:M	One user can generate many audit log entries.
users	report_logs	1:M	One user can generate many report log entries.

11.0 Normalization Evidence

The design is normalized to at least Third Normal Form. The normalization goal is to prevent repeating groups, reduce duplicate data, avoid update anomalies and keep reports dependable. The design also uses lookup tables and associative tables where needed to remove many to many relationships.

Table 31 Normalization Summary

Normal Form	Evidence in the Design
First Normal Form	Each table stores atomic values. Repeating groups such as multiple suppliers for one item are not stored in one field. They are placed in supplier_items. Multiple items in one request are stored in requisition_lines.
Second Normal Form	Tables use single primary keys or associative tables with their own key. Non-key attributes depend on the whole identifier. For example, requisition line quantities depend on the requisition line, not only the requisition.
Third Normal Form	Non-key attributes do not depend on other non-key attributes. Supplier contact information is stored in suppliers, role names are stored in roles, department names are stored in departments and status names are stored in asset_statuses.
BCNF Consideration	Natural identifiers such as username, item_code, asset_tag, serial_number and batch_code are constrained as unique to prevent contradictions between surrogate keys and business identifiers.
Many to Many Removal	Supplier to item is resolved through supplier_items. Requisition to item is resolved

Normal Form	Evidence in the Design
	through requisition_lines.
Derived Data Control	Reports such as low stock and stock movement summaries should be generated through queries or views rather than manually duplicated report tables.

11.1 Sample Functional Dependency Reasoning

The table inventory_items follows the dependency item_id determines item_code, item_name, category_id, uom_id, item_type_id, brand, model and reorder_level. Category details are not stored inside inventory_items because category_id determines category_name in inventory_categories. This removes a transitive dependency.

The table asset_instances follows the dependency asset_id determines item_id, asset_tag, serial_number, batch_id, location_id and current status. Batch details such as supplier and purchase reference remain in acquisition_batches. Location details remain in locations. This prevents repeating location and batch data for each asset.

The table stock_transactions stores historical movement facts only. It records what moved, when, from where, to where and who performed the movement. Current stock is kept in stock_balances for fast lookup, but every balance change must be traceable back to transactions.

12.0 Database Constraints and Integrity Rules

Table 32 Database Integrity Rules

Rule Area	Integrity Rule
Primary Key Integrity	Every table has a primary key. The primary key must be unique and not null.
Referential Integrity	Foreign keys must reference existing parent records. For example, users.role_id must reference a valid role.
Unique Business Identifiers	username, item_code, asset_tag, batch_code and serial_number should be unique where provided.
Quantity Validation	stock_balances.quantity on hand and transaction quantities must not be negative.
Protected Deletion	Records linked to transactions should not be deleted. They should be deactivated to preserve history.
Stock Issue Rule	The system must prevent issuing more stock than is available unless an authorized adjustment is being performed.
Asset Movement Rule	Movement of a fixed asset should reference asset_id so that the asset movement history can be traced.
Audit Rule	Audit logs should be append only and should capture sensitive actions such as login, item update, stock issue and stock adjustment.
Password Rule	users.password hash must store a hash, not a plain text password.
Report Rule	Dashboard reports should be generated from controlled queries or views rather than hand edited report rows.

13.0 Report Support Mapping

Table 33 Report Support Mapping

Required Report	Main Tables Used	Report Logic
Computers in Mechatronics	inventory_items, asset_instances, departments, locations, asset_statuses	Filter category as ICT Equipment and department or location as Mechatronics. Include brand, model, specification, serial number, purchase year, batch and status.
Faulty Assets	asset_instances, asset_statuses, inventory_items, departments, locations	Filter asset_statuses.status_name as Faulty or Under Maintenance.
Low Stock Items	inventory_items, stock_balances, locations, departments, campuses	Compare stock_balances.quantity_on_hand against reorder level or reorder override.
Supplier Transaction History	suppliers, stock_transactions, inventory_items, acquisition_batches	Filter by supplier and transaction type Stock In.
Monthly Stock In and Stock Out	stock_transactions, transaction_types, inventory_items	Group transactions by month, year and transaction type.
Acquisition Batch Status	acquisition_batches, asset_instances, asset_statuses, suppliers	Count assets in a batch by status to show operational, faulty, retired or disposed.
Department Inventory Report	departments, locations, asset_instances, stock_balances, inventory_items	List assets and consumable stock held by department.
User Activity Audit	audit_logs, users, roles	Filter user actions by date, module and affected record.

14.0 Security, Audit and Access Control Design

Database security supports the role based access requirements defined in the SRS. The database stores roles and users, while the application layer enforces permissions. The database also preserves audit logs so that important actions can be reviewed by authorized personnel.

Table 34 Security and Audit Mapping

Security Area	Database Support
Authentication	users stores username, password_hash and account_status.
Authorization	roles links users to their access level.
Department Accountability	users.department_id and asset_instances.department_id support departmental ownership.
Transaction Accountability	stock_transactions.performed_by_user_id records the user who performed the movement.
Approval Accountability	stock_transactions.approved_by_user_id and requisitions.approved_by_user_id store approving users.
Maintenance Accountability	maintenance_records stores reporter and technician users.
Audit Trail	audit_logs stores action, table name, record id, timestamp and value changes where practical.

Security Area	Database Support
Report Accountability	report_logs records reports generated by users where required.

15.0 Implementation Notes for MySQL

The physical implementation should use the InnoDB storage engine because it supports transactions and foreign keys. Character encoding should be utf8mb4 to support a wide range of characters in names and descriptions. Indexes should be added to columns commonly used in searching, filtering and joining.

Table 35 Recommended Indexes

Table	Recommended Indexes
users	username, role_id, department_id
departments	campus_id, department_name
locations	campus_id, department_id, location_name
inventory_items	item_code, item_name, category_id, item_type_id
suppliers	supplier_name, phone
asset_instances	asset_tag, serial_number, item_id, location_id, department_id, asset_status_id, purchase_year
stock_balances	item_id, location_id
stock_transactions	transaction_date, transaction_type_id, item_id, asset_id, supplier_id, performed_by_user_id
requisitions	request_date, requester_user_id, department_id, status
maintenance_records	asset_id, maintenance_status, reported_date
audit_logs	user_id, table_name, created_at
report_logs	user_id, report_name, generated_at

Triggers may be used carefully to update stock_balances after stock_transactions, but the same logic can also be implemented in the application service layer. If triggers are used, they must be documented and tested because hidden database logic can make troubleshooting difficult for student developers.

The design should avoid storing passwords, reports or file uploads directly inside normal text fields where paths and hashes are more appropriate. Item images can be stored as file paths, while the actual files are kept in a protected upload directory.

16.0 Conclusion

This document has presented the database design for the proposed UIRI Inventory Management System. The design separates master data, inventory catalogue records, asset instances, stock balances, stock transactions, requisitions, maintenance records, audit logs and report logs. It supports both Nakawa and Namanve campuses and allows UIRI to track consumable stock and fixed assets in a controlled way.

The next stage after this database design is to review the ERD and data dictionary for approval. Once approved, the design can be converted into SQL table definitions, followed by implementation of the authentication, inventory, stock transaction, supplier, reporting and audit modules.

17.0 References

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